

Table 1: Differentiating augmented, virtual, and mixed reality

	Augmented reality (AR)	Virtual reality (VR)	Mixed reality (MR)
Purpose	Enhancing real-world experiences with digital information (e.g., navigation, education, entertainment)	Creating immersive, simulated environments for various applications (e.g., gaming, training, simulations)	Integrating digital content into real-world environments for interactive experiences (e.g., training, design)
Real-world interaction	Users remain aware of the real world	Users are isolated from the physical world	Real-world interaction is possible
Experience	20% virtual and 80% real	80% virtual and 20% real	40% virtual and 55% real
Gadgets required	Smartphone	VR headset	HoloLens
Control of the senses	Partially controlled	Fully controlled	Partially controlled
Degree of immersion	Moderate	High, as users are fully immersed in a virtual environment	Variable immersion
Use cases	Retail (e.g., virtual try-on), navigation, education (e.g., interactive learning), and gaming with real-world elements (e.g., Pokemon Go)	Gaming, simulations (e.g., flight simulators), training (e.g., medical simulations), and entertainment (e.g., VR movies)	Training and simulations (e.g., industrial training), design and visualisation (e.g., architecture), and interactive experiences (e.g., interactive museums)



Figure 2: Market trends and leading companies in the AR, VR, and MR ecosystem

How AR, VR, and MR are redefining industries

AR, VR, and MR provide immersive training environments that simulate real-world scenarios, allowing for more effective and engaging training programmes. These technologies enable hands-on learning experiences without the need for physical equipment or

spaces, reducing costs, and increasing accessibility. Some of the major industrial applications of AR, VR, and MR are listed below.

Healthcare and medical training:

In healthcare, AR, VR, and MR technologies are transforming patient care, medical training, and research. Some significant applications of these

technologies in healthcare are:

- Medical training and education
- Virtual environments for physical therapy
- Patient care and telemedicine
- Surgical planning and navigation

Retail: In retail, AR, VR, and MR technologies enhance customer experiences, streamline operations, and